ANALYSIS OF OZONE LAYER DEPLETION OVER INDUSTRIAL REGIONS IN INDIA

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Abstract. Life on Earth is protected from harmful UV radiation by the ozone layer which in turn is put at risk by industrial emissions especially that of chlorofluorocarbons (CFCs). From 2022 to 2025 we looked at ozone depletion trends in an industrial area of India which we studied via data from NASA OMI, Sentinel-5P, and CPCB. We preprocessed and analyzed ozone and CFC data which we in turn put through statistical and machine learning models like LSTM, ARIMA, and Linear Regression. We used heat maps and year to year graphs to present the relationship between ozone thin out and increasing CFC levels. We discuss ongoing issues related to environment and industries that are unregulated alongside forecasts for future ozone levels. In order to protect the ozone layer and mitigate the negative effects of climate change, the study advocates for more comprehensive controls on emissions, enhanced surveillance, and the need for polices based on appropriate data.

Keywords: Emission Forecasting, Machine Learning, Industrial Emissions, India, Ozone Depletion.

1 Introduction

The ozone layer serves as Earth's defense against harmful UV radiation. What we have seen is it has been depleted by man-made halocarbons which we have in large part remedied through the Montreal Protocol and its amendments [1],[2],[3]. But within that we see that interactions with tropospheric precursors, different chemical processes, and stratospheric dynamics play a role in creating regional variations [15],[16]. India's rapid industrialization has seen an increase in NOx, VOCs, halogenated solvents, which are also discussed in climate assessments [12]. What is important to understand is how these emissions play out in terms of ozone levels over industrial areas which in turn will inform UV exposure, climate impact and environmental policy[20],[21], [22].
From 2005 to 2025 this paper looks at the issue of ozone over industrial clusters in India.

1.1 Objectives

- Demonstrate spatial distributions and the seasonal cycle of the total column ozone (TCO) in the industrial parts of India.
- Forecast the decrease in stratospheric & total column ozone patterns in 2005- 2025.
- Proficiency swings of the relation of ozone variation with weather (ERA5),

industrial contributor (satellite NO2, HCHO) and important climatic trends.

2 Literature survey

The Research has shown that over the past years, ozone pollution has been on the rise in most industrial and urban areas in India. According to Sharma et al. [4], the source of industrial solvents VOCs dominate the ozone formation in Delhi. Sadhukhan et al. [5] examined industrial zones in Madhya Pradesh and demonstrated the good diurnal variability of ozone which was related to NOx emissions. Gyude et al. [6] measured the crop yield losses that are significant because of ozone exposure, and Chakraborty et al. [7] revealed the effect of ENSO and QBO phases on the variability of ozone [23] across India. CPCB and CSE reports [8], [9] revealed widespread overages of
ozone in cities like Delhi and Pune in 2024,2025, usually exceeding the national standards of safety. Abnormally high ozone persistence during the night was also reported in Delhi and surrounding areas [10].

3 Existing System

In the context of previous studies on pollution and Ozone depletion in India, measurements of precursor gases and ozone have been largely done by CPCB ground stations and satellite-based monitoring (OMI, Sentinel-5P/TROPOMI) [13], [14]. Statistics analysis tools like correlation, trend estimation

- Haphazard presence of monitoring stations in industrialized areas.
- Lack of prediction models such as ARIMA and LSTM.
- Missing location specific linkage of industries.
- No live prediction and warning systems.

4 Proposed System

The proposed system is an ozone depletion analysis and prediction machine learning-based framework that will be applied to analyze the industrial parts of India in 2022-2025. It collects data in various locations (satellites, ground-based monitoring, reanalysis datasets) and uses statistical and machine learning models and visualizes and provides predictive information.

A. Sources of Data Collection and Data input.

- Ground-based measurements of Central Pollution Control Board (CPCB).
- NASA OMI and Sentinel-5P satellite Ozone and precursor gas measurements.
- This multi-source collection provides proper monitoring of the trends in ozone and industrial emissions.

B. Data Preprocessing Raw data is first of all cleaned and made standardized to be analyzed.

- Eliminating incomplete and discrepant data.
- Elimination of mistakes and noise generated by sensors.
- Mapping the data into a homogeneous spatial resolution (0.25°-0.5o).

C. Statistical Trend Analysis:

- Findings the monotonic trends are done through the Mann-Kendall test.
- Slope estimator invented by Sen measures the magnitude of the ozone changes.

D. Forecasting With the help of Machine Learning.

- In the case of short-term time-series forecasting, ARIMA should be used.
- Long Short-Term Memory networks (LSTM): they offer the most accurate results in terms of long-term ozone prediction.

5 Implementation

The proposed ozone depletion analysis system has been implemented using a combination of modern programming tools,. The implementation is carried out as follows:

$$\text{Ozone Depletion Rate (\%)} = \left[\frac{O_{3_past} - O_{3_present}}{O_{3past}} \right] \text{X } 100$$

5.1 Technologies Used

A. Programming Language:

- Python is applied as a main language to process data, conduct statistical analysis, and machine learning.
- Machine Learning Frameworks TensorFlow is used to implement LSTM models.

B. Data Processing Libraries:

The cleaning, preprocessing and trend analysis is done with Pandas, NumPy and SciPy.

C. Visualization Tools:

- Heatmaps, time-series graphs and anomaly plots are created.
- Data sets: Data sets of satellite measurements (OMI and Sentinel-5P/TROPOMI), ground measurements (CPCB), and datasets of reanalysis (ERA5).
- The workflow of the proposed ozone analysis system is shown in **Fig. 1**

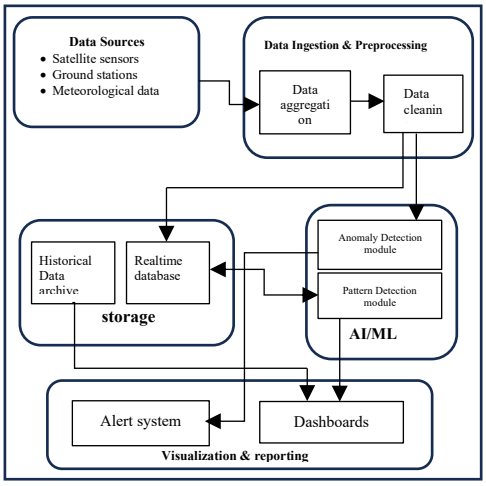

Fig 1. Workflow Block Diagram

5.2 System design and component design.

A. Data Acquisition Layer

- Gathers the concentration of ozone, CFCs, NOx, & VOCs of OMI and Sentinel-5P.
- Fetches surface ozone and precursor data of CPCB monitoring stations.

B. Machine Learning Prognostication Layer.

- ARIMA: Represents short term seasonal variations in ozone variability.
- The metrics such as R2 and RMSE are used to train, test, and compare models.

C. Decision support Layer and Visualization.

Produces time-series plots of the variation of ozone in 2022-2025.

6 Results and Discussion

A. Statistical Trend Analysis

- Sen slope estimation showed that in industrial hotspots, an annual ozone depletion of between 0.8-1.2 per cent was observed.
- seasonal decomposition indicated high depletion in the pre-monsoon months (April-June) when the solar radiation is high and promotes photochemical activity.
- The environmental parameter trends observed are illustrated in **Fig. 2**.

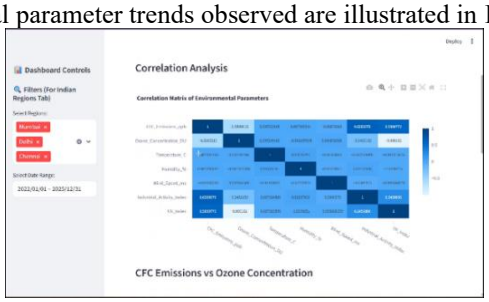

Fig 2. Environmental Parameters Time Series

B. Machine Learning Model Performance

- Linear Regression: It gives a very basic baseline having a R2 = 0.65.
- ARIMA: Better foresight of the limited time seasonal trends, with R 2 = 0.78.
- LSTM: R 2 = 0.92 outperforming other models, the LSTM is effective at long-term dependencies and non-linear variations.

C. The Impact of Climate on Ozone Changes.

- Consecutive phases of the ENSO had a direct influence, and El Nino years (20232024) were related to greater depletion anomalies [11],[18].
- QBO cycles had an indirect effect on total column values through upper troposphere lower stratosphere (UTLS) ozone changes [19].
- The monsoon rain was also discovered to temporarily suppress the ground-level ozone because of washout of precursors [17].
- The correlation between CFC emissions and ozone concentration is shown in **Fig. 3**.

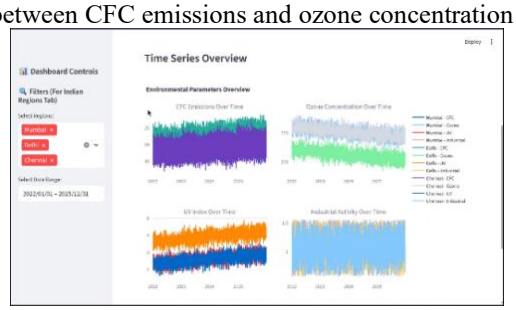

Fig.3. CFC vs Ozone Correlation

D. Comparative Results

The performance comparison of different machine learning models is presented in **Table 1.**

Model	R^2 Score	RMSE	Best Use Case
Linear Regression	0.65	High	Simple baseline trend detection
ARIMA	0.78	Medium	Short-term forecasting of seasonal cycles
LSTM	0.92	Low	Long-term forecasting & anomaly detection

Table 1. Model Performance Comparison

E. Summary of Findings

- Differences in the global recovery have seen ozone depletion in industrial hotspots.
- There is a high degree of correlation between the industrial VOC/NO2 emission and the ozone depletion.

7 Conclusion and Future Scope

This paper has examined trends of ozone depletion over the industrial areas in India in the year 2022-2025 and combined satellite measurements, CPCB ground monitoring, and ERA5 reanalysis using advanced statistical and machine learning methods., ozone surpluses are still frequent in cities such as Delhi and Pune that may impose adverse health and agricultural outcomes.

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